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Mapping targetable sites on the human surfaceome for the design of novel binders

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The human cell surfaceome, integral to cell communication and disease mechanisms, presents a prime target for therapeutic intervention. De novo protein binder design against these cell surface proteins offers a promising yet underexplored strategy for drug development. However, the vast search space and limited data on natural or competitive binders have historically limited experimental success. In this study, we systematically analyzed the entire human surfaceome, identifying approximately 4,500 targetable sites and introducing potential binding seeds for initiating protein design applications. To validate these seeds, we implemented two experimental approaches (protein scaffolding and peptide cyclization) on three representative targets (FGFR2, IFNAR2, and HER3). Our results revealed a high success rate, showing that seeds provide valuable starting points for binder design against our identified targetable sites, as well as the need for constant improvements of computational protein design pipelines utilizing machine learning and physics-based methods. Additionally, we present SURFACE-Bind, an interactive database offering open access to all generated data. The high-throughput computational design methods and target-specific binder seeds established here pave the way for a new generation of targeted therapeutics for the human surfaceome.

de novo protein design | geometric deep learning | human surfaceome | protein–protein and protein-peptide interactions | mapping of targetable binding sites

Significance

The human cell surfaceome presents a vast yet underexplored landscape for therapeutic intervention using protein-based binders. In this study, we systematically mapped 4,500 targetable sites across all human cell surface proteins and identified structurally validated binder seeds as starting points for de novo protein design. Experimental validation on three targets confirms the potential of these seeds as precise starting points for diverse binder design strategies. We also introduce SURFACE-Bind, an open-access database that enables researchers to explore these targetable sites and facilitate future computational and experimental protein design efforts. By providing a comprehensive database of targetable binding sites along with their corresponding binder seeds, this work advances computational protein design and lays the foundation for developing new protein-based therapeutics.

The diverse array of human cell surface proteins, known as the human surfaceome (1), plays a crucial role in cell-to-cell communication and the transmission of signals from the outside to the inside of cells. Because these surface proteins are easily accessible, the surfaceome represents a rich resource for drug targeting and has emerged as a promising avenue for therapeutic development. It is estimated that the human surfaceome includes around 2,886 proteins, categorized into different families and subfamilies (2). While some of these proteins, such as programmed death-1 (PD-1) (3), human epidermal growth factor receptor 2 (HER2) (4), and epidermal growth factor receptor (EGFR) (5), have been extensively targeted in protein design efforts, many others remain poorly understood or unexplored.

Computational design of de novo protein binders from scratch poses several challenges and requires innovative strategies to unravel the complexities of molecular interactions. Recent advances in deep learning have significantly enhanced protein design approaches (6), allowing for precise structural accuracy and higher experimental success rates (7–9). However, this remains a difficult task, particularly in cases where there are multiple potential binding sites on a target protein and limited or no available data on natural protein binders. To address these challenges, we recently introduced MaSIF-seed (10), a geometric deep-learning framework designed specifically for protein surfaces. This framework generates "fingerprints" that capture both the geometric and chemical features essential for protein–protein interactions (PPIs). Our findings demonstrated that these fingerprints provide insights into the structural complementarity of molecular complexes and also present further possibilities for designing novel protein binders.

Building on the significance of the human surfaceome and the success of the MaSIF framework, we have now systematically evaluated all proteins within the human surfaceome, identifying approximately 4,500 potential binding sites across 2,886 proteins, which could be leveraged in future protein design studies. We further assessed the binding propensity of each site based on a combination of geometric and chemical scores in its unbound state. These predictions assess the feasibility of targeting each site through pairwise docking of 640,000 seed structures. This analysis enabled us to propose a set of potential binding seeds together with seed-bound-state scores that can be used to advance protein design efforts. To illustrate the potential of these seeds, we applied multiple design approaches to three cell surface targets, designing both protein- and peptide-based binders, including our published MaSIF-seed

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approach (10), an optimized version that incorporates machine learning tools, and a diffusion-based peptide design pipeline (Cyclotide). While the affinities of some designs could be further improved, our results demonstrate that the target sites together with the initial seeds serve as effective starting points for binder designs in different molecular formats. By establishing a comprehensive database of these seeds targeting predicted binding sites in the human surfaceome, our approach significantly enhances the scale and efficiency of peptide and protein design efforts, paving the way for new opportunities in developing therapeutics. All data generated in this study are freely accessible through the interactive SURFACE-Bind database (https://surface-bind.inria.fr/) (11), enabling users to explore target proteins of interest and to further design protein binders by utilizing the detected seeds.

Results

Computational Analysis of the Human Surfaceome. We performed a large-scale comprehensive analysis of the targetability of the human surfaceome. As indicated in Fig. 1*A*, we targeted all proteins from the human in silico surfaceome database, SURFY, which encompasses 2,886 proteins that are either experimentally determined or predicted to be surface proteins (2). These proteins are classified functionally into six main families: receptors, transporters, enzymes, miscellaneous, unclassified, and unmatched. For each protein entry, the corresponding AlphaFold2 (AF2) (12) model was obtained and filtered based on the predicted local distance difference test (pLDDT) (see *Methods* section). A large-scale site prediction approach was carried out via the molecular surface interaction fingerprints (MaSIF-site) framework (13), mapping the surface of each protein into a point cloud and predicting which surface points are more likely to interact with other proteins. To define the binding sites and to distinguish between the boundaries of two binding sites in close proximity, these points were subsequently clustered using density-based spatial clustering of applications with noise (DBSCAN). As an end result, we predicted a total number of 4,500 target sites excluding the transmembrane (TM) domains. The average number of sites per family and subfamily is shown in Fig. 1*B*, where the G protein-coupled receptor (GPCR) and solute carrier family (SLC) obtained the lowest number of sites,

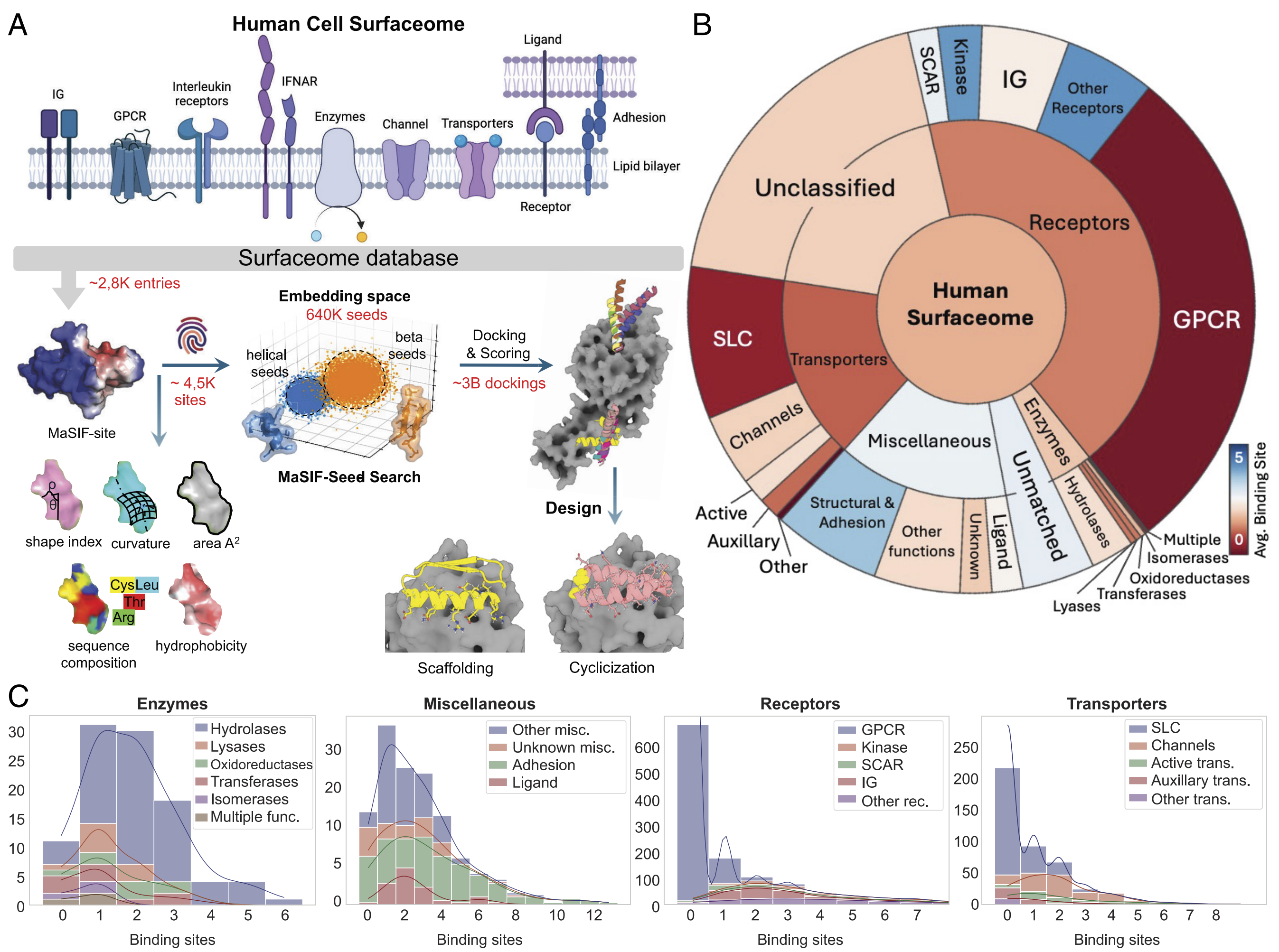

Fig. 1. Overview of computational workflow and targetable sites in the human surfaceome. (*A*) Schematic overview of the computational workflow. The Surfaceome structures were gathered from the SURFY database, comprising 2,886 entries (2) High propensity protein binding sites were predicted using MaSIF-site (13), resulting in a total of 4,500 binding sites. These binding sites were then targeted by a library of 640,000 protein fragments based on the MaSIF-seed algorithm (10) and followed by pairwise docking with the evaluation of around 3 billion interactions. Subsequently, two protein design approaches, including scaffolding and cyclic peptide design, were then applied on representative proteins to evaluate the quality of seeds under experimental settings. (*B*) Average predicted binding sites per family. Segment size corresponds to the number of proteins in each family, with GPCRs having the largest protein count but the fewest predicted binding sites, primarily located within the transmembrane (TM) domain. (*C*) Binding site frequency on the four main protein families. Each subfamily is shown in a different color. Miscellaneous and transporter main classes have several members with multiple predicted binding interfaces, where some members in adhesion proteins contain more than 12 binding interfaces.

while Kinase and Other Receptors have an average of six binding sites per protein. The frequency of detected sites per family and subfamily is depicted in Fig. 1*C*.

To evaluate the targetability of the predicted sites, we computed two sets of scores, including unbound- and seed-bound-state scores. The unbound scores are composed of geometric and chemical features such as shape index, surface curvature (concave, convex, and flat), estimated binding interface area (A^2), sequence composition, solvent accessible surface area (SASA), and the hydrophobicity scale (14). The distribution of unbound-state features across each family and subfamily is shown in *SI Appendix*, Figs. S1 and S2. Analyzing the sequence composition of the predicted sites, we noted that leucine, valine, and serine are the most frequent residues in the predicted binding sites (Fig. 2*A*), in line with the typical hydrophobic character found in PPI sites.

Next, in order to calculate the bound-state scores, we targeted these sites by a library of 640,000 small helical and beta strand fragments (seeds) derived from the Protein Data Bank (PDB). We performed and evaluated nearly 3 billion pairwise dockings, resulting in bound-state scores and a selection of high-quality seeds for each target site with the highest complementarity matching score (10) (depicted in Fig. 2*B* and *SI Appendix*, Fig. S1). MaSIF-seed was also capable of identifying diverse structural seeds for targeting predicted PPI sites, thereby providing an opportunity for subsequent binder generation. On average, receptors within the Ig family exhibit the highest number of matching seeds. As depicted by three examples, the detected PPI sites have also been found to overlap with the natural binder (Fig. 2*C*), which could potentially disrupt the biological function of the protein.

In addition, we investigated the receptor–receptor interactions which encode another layer of biologically and therapeutically

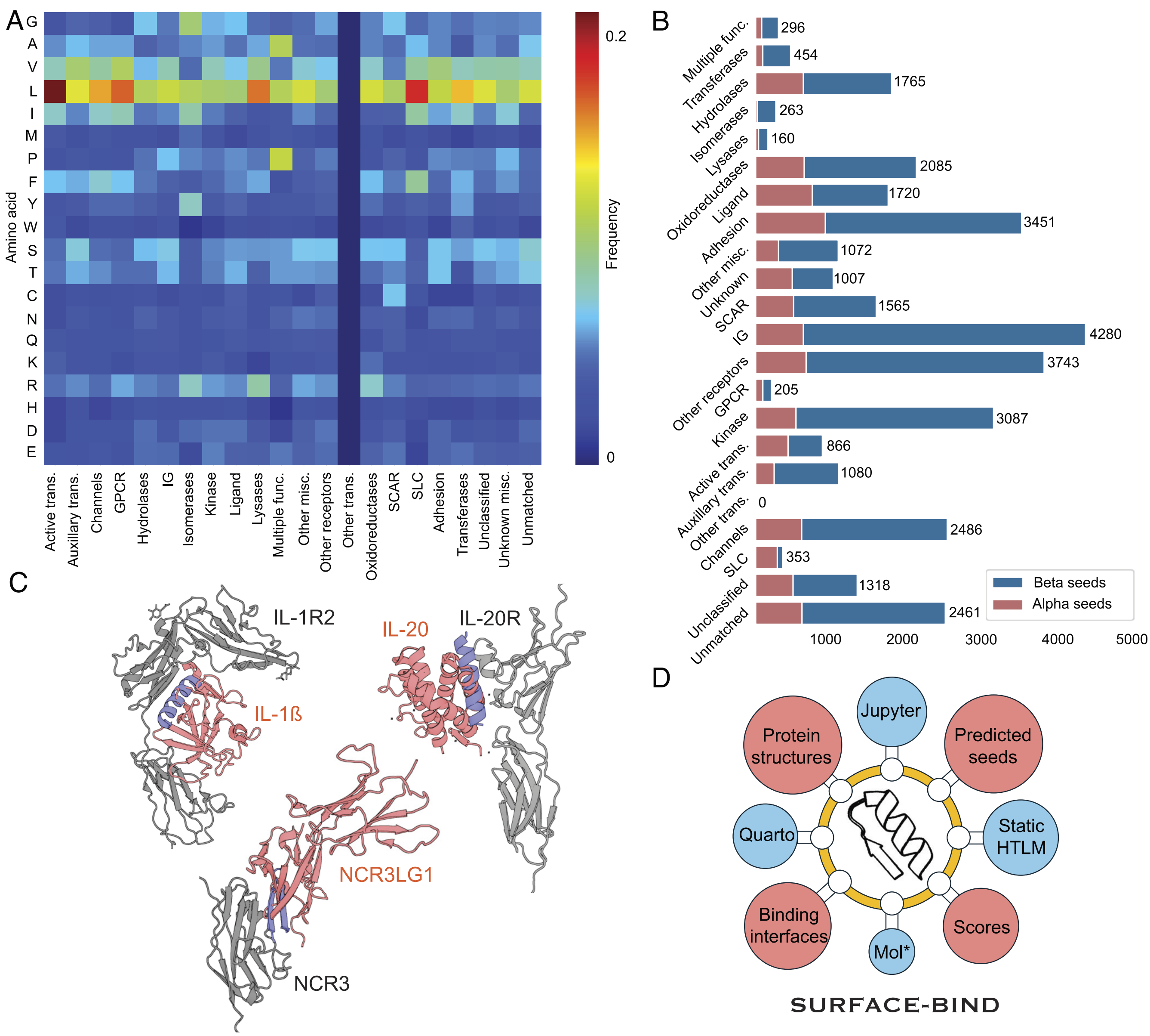

Fig. 2. Mapping of the targetable human surfaceome. (*A*) The amino acid frequency on the predicted binding sites for all subfamilies. Leucine is the most frequent amino acid found in the predicted binding sites across all main classes and specifically in the transporters family. Valine and serine are the second and third frequent amino acids, respectively. (*B*) Average number of alpha and beta seeds per subfamily. The number of detected alpha seeds is lower than the number of beta seeds, primarily due to the difference in the initial size of their databases. Although families with a higher number of binding sites obtained a higher number of seeds, the Ig subfamily showed a significant bind-ability to beta seeds. (*C*) Examples include entries with natural binders (gray and red) and detected inhibitory seeds (blue) that disrupt the interaction. (*D*) The SURFACE-Bind database provides the data openly available through static and interactive HTML pages.

relevant interfaces. Receptor kinases and other surface receptors frequently form homo- and heterodimers or to higher-order assemblies, and targeting these interfaces may provide unique opportunities to mediate receptor signaling (15–17). We analyzed all receptor entries with available crystal structures in the PDB (248 entries) and evaluated whether predicted binding sites coincided with experimentally determined receptor–receptor interfaces. We identified 57 such interactions in the PDB. In 54 of these cases, our predictions correctly captured the binding regions located at the receptor–receptor interface or, in a few instances, at interfaces mediated by an intermediate protein. Eight representative examples are shown in *SI Appendix*, Figs. S3 and S4, and the complete list of receptor–receptor interactions, including UniProt and PDB identifiers, is provided in Dataset S1.

We further benchmarked our predictions against known antibody epitopes, as antibodies often define physiologically accessible sites and provide opportunities for competition assays with designed binders. To this end, we cross-referenced all 2,886 surfaceome entries against the structural antibody database (SAbDab) (18) and identified 147 proteins with antibody or nanobody complexes. Among these, 52 proteins had antibody epitopes that overlapped with or were in close proximity to our predicted binding sites (10 to 12 Å). Eight representative examples are illustrated in *SI Appendix*, Fig. S5, with full details provided in Dataset S2.

Finally, we examined the role of posttranslational modifications (PTMs), with a particular focus on glycosylation, which is known to modulate surface accessibility and may hinder or orientate binder engagement. We systematically queried all predicted binding sites against UniProt glycosylation annotations and found that 616 surfaceome entries contained at least one glycan in proximity to a predicted binding site. At the site level, ~1,070 out of ~4,500 predicted sites were affected, highlighting the widespread influence of glycosylation. While only 18 entries were directly impacted when considering the binding-site center residue alone, including all neighboring residues revealed a much broader potential for interference. The full set of glycosylation annotations, including residue positions and glycan types, is available in Dataset S3.

In order to make the data freely available to the community, we created an interactive database called SURFACE-Bind (Fig. 2*D*) (11). This database comprises static HTML pages for individual entries, each associated with a specific UniProt ID. Each page contains detailed information on the protein structure, predicted binding interfaces, detected seeds, and several visual plots showcasing the distribution of chemical properties for both the binding interfaces and the corresponding seeds. All associated files can be conveniently downloaded in compressed format at the bottom of each entry page, providing easy access to specific data.

De Novo Designed Binders against Cell Surface Receptors. After analyzing the entire human surfaceome and its propensity to be involved in PPIs, we sought to generate de novo protein designs against three major cell surface receptors: fibroblast growth factor receptor 2 (FGFR2), human epidermal growth factor receptor 3 (HER3), and interferon alpha/beta receptor 2 (IFNAR2). MaSIF-site identified one site with a high interface propensity for each targeted structure (Fig. 3*A*). All putative interfaces were situated at or near previously identified binding sites (*SI Appendix*, Fig. S6). Using the MaSIF-seed-search pipeline (10), we computationally generated and selected 2,000 protein designs against each target. Briefly, top-ranking seeds were refined with the Rosetta modeling suite (19) and grafted on recipient scaffold protein, generating about 2,000 designs for each target (rd1) and sampling a wide range of folds and topologies (Fig. 3*B*).

All computational designs were screened by yeast surface display, and after two rounds of sorting, we observed clear binding populations to their respective targets, which were selected for deep sequencing. We identified 10 designs for FGFR2, three designs for HER3, and one design for IFNAR2 for which deep sequencing showed an enriched binding profile, confirmed using flow cytometry (*SI Appendix*, Table S1). Half of the designs showed the predicted binding mode as assessed by mutagenesis at the designed interface (four for FGFR2, one for IFNAR2, and two for HER3) (*SI Appendix*, Fig. S6 and Table S1).

Upon expression in *Escherichia coli* (*E. coli*) (Fig. 3*C*) and subsequent purification, these candidate binders were found to adopt dimeric or tetrameric oligomeric states in solution. Evaluation using surface plasmon resonance (SPR) revealed a binding affinity ranging from high nanomolar to low micromolar, as well as a high thermal stability (Fig. 3*D* and *SI Appendix*, Fig. S7). Importantly, competition assays showed the absence of binding in the presence of a known competitor, confirming that the correct target interfaces are engaged by the different designs (Fig. 3*E* and *SI Appendix*, Fig S6 and Table S1).

Of note, one of our designs, H_rd1_309, only expressed in the presence of a soluble fusion protein (Maltose Binding Protein, MBP) and demonstrated a binding affinity higher than double-digit μM, most probably because of improper folding (*SI Appendix*, Fig. S8). We therefore sought to improve its stability and expressibility by using ProteinMPNN (20), with fixed binder interface residues and filtering the designed sequences with AF2 prediction metrics to retain one candidate (*SI Appendix*, Fig. S8). The prediction of the optimized design done by AF2 multimer (21) also showed perfect alignment with the original model (1.64 Å C_{α}-rmsd). Finally, the optimized H_rd1_309 expressed in high yield without soluble fusion protein and showed increased affinity on SPR (K_D = 8.7 μM). This result, together with similar findings from other groups (8, 9, 22) underscores the need for an optimized pipeline that leverages state-of-the-art deep learning tools to improve and select designed protein binders.

Enhancing Experimental Successes with ML Tools. Following the observed limited experimental success in terms of binder design success rate (for instance 4/2,000 binders for FGFR) and the binders' challenging biophysical properties (e.g., low expression and stabilities), we set out to improve the MaSIF-seed design pipeline by leveraging the machine learning toolbox including ProteinMPNN (20) and AF2. The previously seen shortcomings could have resulted from the designs failing to a) fold properly and/or b) bind the targeted site effectively. Within the MaSIF-seed pipeline, the bulk of the mutations were performed to design the protein interaction site, while the remainder of the scaffold was largely untouched.

Hence, we sought to optimize the MaSIF-seed pipeline using ProteinMPNN and assess their in-silico foldability and interface quality using AF2 (Fig. 4 *A* and *B* and *SI Appendix*, Table S2), as a strategy to enhance their biochemical behavior. In the first stage of optimization, ProteinMPNN was used to optimize the designed sequences in two settings: i) noninterface residues and ii) full redesign of the binders, in an attempt to improve their affinity. The in silico foldability of the ProteinMPNN-generated sequences was determined employing AF2 predictions. The designs were filtered based on pLDDT and C_{α} rmsd to the original model. The filtered designs' interface quality was then assessed employing complex prediction using the AF2 monomer model and selecting for candidates with interface predicted template modelling (ipTM) score ≧ 0.7 (21, 23). Finally, we tested ~180 sequences for each of FGFR2 and IFNAR2, and ~500 sequences for HER3.

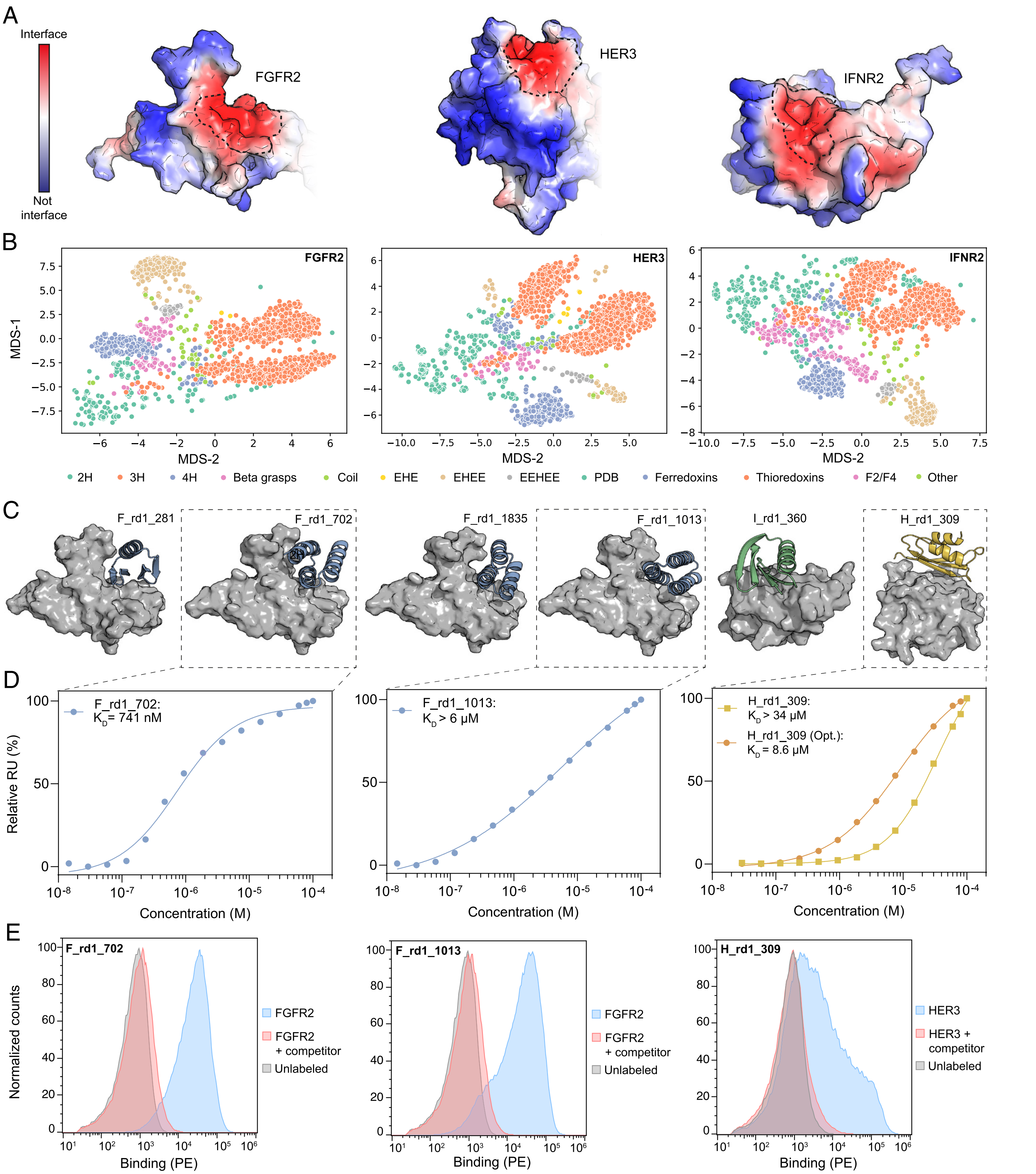

Fig. 3. Generation of de novo designed binders against surface receptors. (*A*) MaSIF-site interface predictions for the three protein targets. Red color indicates a high interface propensity, while blue color indicates a low propensity. (*B*) Fold diversity of the ~2,000 computational designs for the three target proteins mapped using multidimensional scaling (MDS) of pairwise rmsds between all designs. (*C*) Computational models of the six experimentally validated designs. Target proteins are shown in gray, and designs are colored in blue (FGFR2 binder), green (IFNAR2 binder), or yellow (HER3 binder). (*D*) Affinity measurements of three expressing binders performed by SPR. Steady state responses and dissociation constants were plotted using a nonlinear four-parameter curve fitting analysis. (*E*) Histograms of the binding signal (PE; phycoerythrin) measured by flow cytometry on yeast displaying designed binders. Yeast were labeled with their respective target protein in free form (blue) or preblocked with a competitor protein (red; 7NIJ for FGFR2 and RG7116 for HER3) or unlabeled (gray).

The optimized designs (rd2) were screened using yeast display, and after two rounds of sorting, yeast cells exhibiting binding signals to their target proteins were selected for deep sequencing. A subset of designs, selected based on enrichments derived from deep sequencing data, was expressed in *E. coli*. All designs were readily expressed except for F_rd2_215 (*SI Appendix*, Table S1).

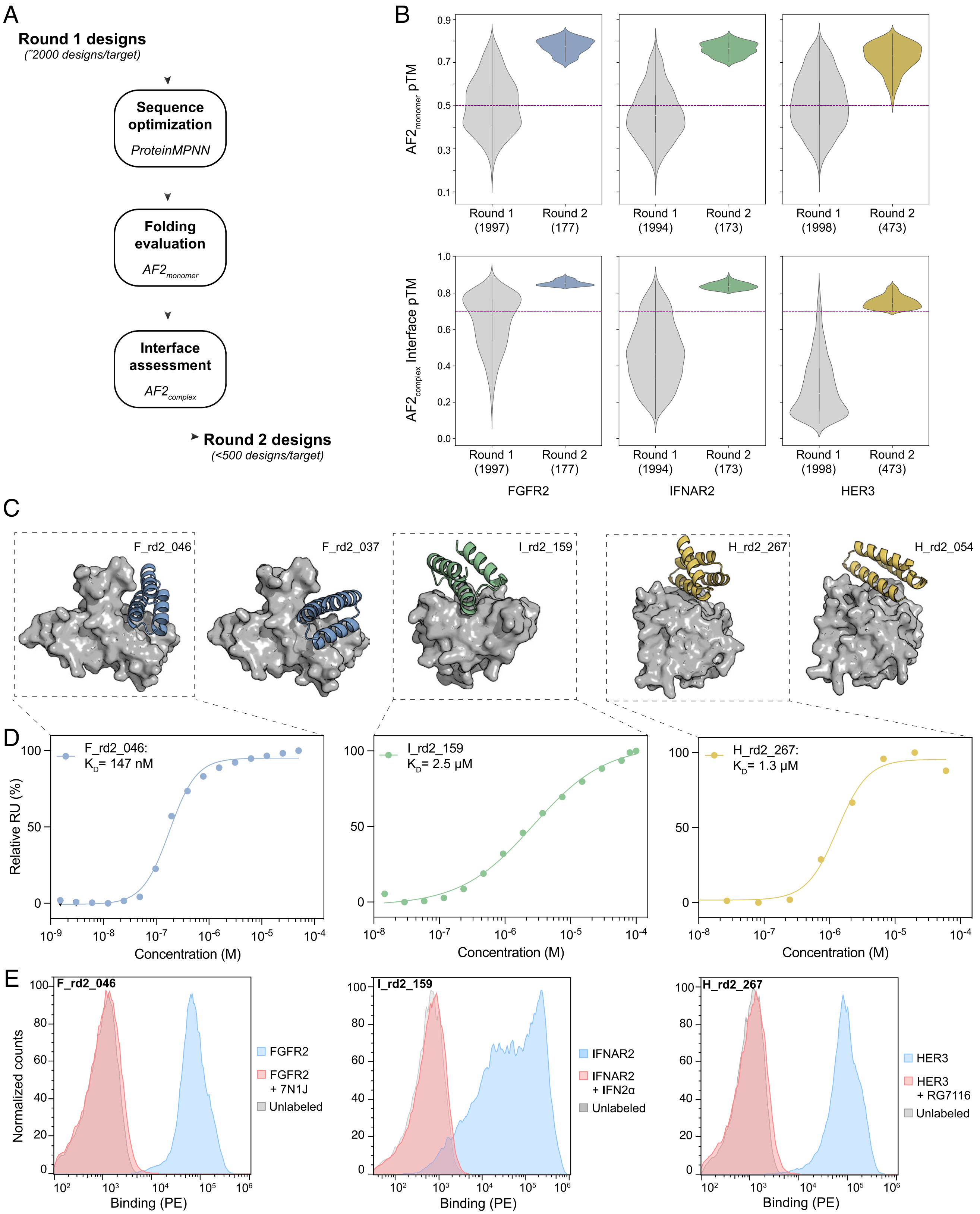

Fig. 4. Improved design pipeline and experimental characterization of optimized designs. (*A*) An enhanced computational pipeline including ProteinMPNN for interface design and/or scaffold optimization and AF2 for monomer folding and complex prediction (*B*) Comparison between designs before and after optimization for AF2 monomer predicted template modelling score (pTM) (*Top*, dashed line set at pTM = 0.5) and AF2 multimer ipTM (*Bottom*, dashed line set at ipTM = 0.7). (*C*) Computational models of five experimentally validated optimized designs. Target proteins are shown in gray and protein designs are colored in blue (FGFR2 binder), green (IFNAR2 binder), and yellow (HER3 binder). (*D*) Affinity measurements of three binders performed by SPR. Steady state responses and dissociation constants were plotted using a nonlinear four-parameter curve fitting analysis. (*E*) Histograms of the binding signal (PE, phycoerythrin) measured by flow cytometry on yeast displaying designed binders testing the binding specificity with competing reagents. Yeast were labeled with their respective target protein in free form (blue) or preblocked with a competitor protein (red) or unlabeled (gray).

The purified designs were folded, monomeric, and showed high nanomolar to low micromolar affinities against their respective targets, as determined by SPR, namely F_rd2_046 (K_D = 147 nM), I_rd2_159 (K_D > 2.5 μM), and H_rd2_267 (K_D = 1.3 μM) for FGFR2, IFNAR2, and HER3, respectively (Fig. 4 *C* and *D* and *SI Appendix*, Figs. S10 and S11). Interface knockout (KO) mutants and competition assays with known binding reagents were consistent with the designed modes of interaction (Fig. 4*E*). Our results highlight the key role of the optimized pipeline utilizing ProteinMPNN and AF2 in improving the experimental success rates for identifying site-specific binders, with enhancements exceeding 11-fold and 8-fold compared to rd1 designs for FGFR2 and HER3, respectively. Moreover, while the lead rd1 designs targeting IFNAR2 failed at the expression stage, the optimization process successfully produced a low micromolar binder (*SI Appendix*, Table S3).

To assess the contribution of ML tools to the experimental successes of the rd2 designs, we compared the in silico metrics of the designs before and after optimization. The rd2 design set showed a much higher in silico folding propensity as evaluated by AF2 as compared to the rd1 designs (Fig. 4 *B*, *Top* and *SI Appendix*, Fig. S12*A* and Table S4). The key discriminating metric was the interface quality, inferred from the ipTM score, where most of the rd1 designs were below 0.7 (Fig. 4 *B*, *Bottom* and *SI Appendix*, Table S4). Although for FGFR2, 877 of 1,997 designs (44%) passed the ipTM threshold, further filtering with AF2 monomer thresholds to exclude misfolded designs reduced this set to 91 potential binders (4.5%). These findings align with our hypotheses that the lower success rates of rd1 designs were likely due to suboptimal protein scaffolds hindering folding and consequently binding. In silico analysis of employed scaffold foldability indicated that, although the native scaffolds exhibited favorable folding properties (*SI Appendix*, Fig. S12*D*), this was compromised in the rd1 designs (Fig. 4*B*). This decline likely reflects limited scaffolds' mutational tolerance, introduced by Rosetta in the redesign steps, leading to misfolding. Sequence similarity analysis revealed that more than half of the sequence were altered in most rd2 designs compared to their predecessors (*SI Appendix*, Fig. S12*E*).

Of the six lead candidates, two were obtained through the optimization of the scaffold while retaining the original interface, and four resulted from the fully redesigned sequences. Although the rd2 AF2 complex predictions closely aligned with the rd1 models, showing only minor structural deviations on the binders (*SI Appendix*, Fig. S13 *A* and *B*), the rd2 designs demonstrated improved AF2 metrics for both in silico folding and complex interface quality compared to their respective rd1 designs (*SI Appendix*, Fig. S13*C*). These results highlight the capabilities of AF2 to aid in protein design tasks.

Since our designs were built on initial seeds identified via the MaSIF-seed pipeline, we examined how interface residues evolved across the design stages. Seed interfaces were first redesigned in Rosetta to optimize interaction energetics before grafting and further optimization steps in the rd1 design pool. Comparison of the sequences before (MaSIF seed) and after Rosetta redesign (refined seed) revealed that over 50% of interface residues were mutated (*SI Appendix*, Fig. S14). As the refined seed formed the basis of the rd1 designs, we compared its interface with those of the rd2 designs (*SI Appendix*, Fig. S15). Interface conservation across designs ranged from 14.3 to 77.8% (*SI Appendix*, Fig. S15*A*). Most substitutions involved residues with similar physicochemical properties and size, suggesting that the steric and hydrophobic character of the refined seed interfaces was largely preserved. However, nonconservative mutations showed a consistent bias toward increased hydrophilicity. This shift might promote the formation of interface hydrogen bonds, enhance binder folding, and improve both interface quality and binding affinity (*SI Appendix*, Fig. S15 *B* and *C*).

We performed a computational benchmark assessing the success rate of our optimized pipeline in full redesign setting to generate binders that fulfill AF2 filters for monomer foldability (pLDDT ≥ 90, pTM ≥ 0.7, and C_α rmsd ≤ 1.5Å) and interface quality (ipTM ≥ 0.7) compared to de novo binder design using RFdiffusion and ProteinMPNN toward the same sites in all three targets in this study. Our pipeline demonstrated higher computational success rates for two targets (*SI Appendix*, Fig. S16 and Table S5), although further experimental validation will be necessary to assess the performance.

Generation of Peptide Binders from De Novo Proteins. Building on the success of miniprotein binders targeting FGFR2, IFNAR2, and HER3 and to demonstrate the versatility of our binding motifs as starting points for formats beyond protein scaffolds, we expanded our approach to design peptide binders, which offer additional advantages as modality due to the smaller size. Two design strategies were utilized: i) cyclized peptides generated by diffusion-based methods and ii) stabilized PPI motif peptides derived from the lead candidates of the miniprotein binders using rational protein design techniques (Fig. 5*A*).

Starting from the experimentally validated rd1 and rd2 designs, the binding interface motifs were analyzed for their suitability to design cyclized peptides. Defined secondary structural motifs such as helix, helix-turn-helix, and beta-hairpins were amenable to stabilization through disulfide bonds, achieved either by introducing terminal cysteine residues or incorporating disulfide-anchored loops. In some instances, disulfide-stabilized motifs were further embedded into larger cyclized peptides (up to 38 residues in length) featuring two disulfide bonds and employing the cyclotide design pipeline. The cyclized peptides were evaluated in silico for their predicted folding, using AF2 with relative positional encoding adjusted to account for interresidue distances in a cyclic sequence (24), and for the stability of the engineered disulfide bonds, using molecular dynamics (MD) simulations. Additionally, a minimalistic approach was employed, wherein helical interface motifs were extracted and stabilized through the addition of a thiol-reactive linker (Fig. 5*A*). A total of 16 peptide candidates were designed, synthesized, and experimentally tested, resulting in five target-specific binders (Fig. 5*B*).

Twelve of the 16 peptides were soluble in PBS, with four showing folded structures in solution as assessed by circular dichroism (CD) spectroscopy: two peptides (F_p10 and F_p15) in PBS and two (F_p1 and H_p8) in the presence of 20% TFE (2,2,2-Trifluoroethanol) (*SI Appendix*, Fig. S17 and Table S6). Binding of the 12 peptides was tested with SPR against their respective targets (Fig. 5*C*). Target specificity for the designed peptides was determined by SPR against control proteins and by interface KO mutants against their targets (Fig. 5*C* and *SI Appendix*, Fig. S18). We identified five target-specific peptides, three targeting FGFR2 (F_p4, F_p10, and F_p15) and two targeting HER3 (H_p5 and H_p8) (Fig. 5*C* and *SI Appendix*, Fig. S18).

Discussion

In this study, we employed a surface-based geometric deep learning approach (MaSIF-seed) to systematically predict and analyze protein binding sites across the entire human surfaceome. This extensive analysis unveiled approximately 4,500 sites that are potentially targetable by protein-based ligands, opening new routes to modulate the function of these cell-surface proteins. Beyond individual

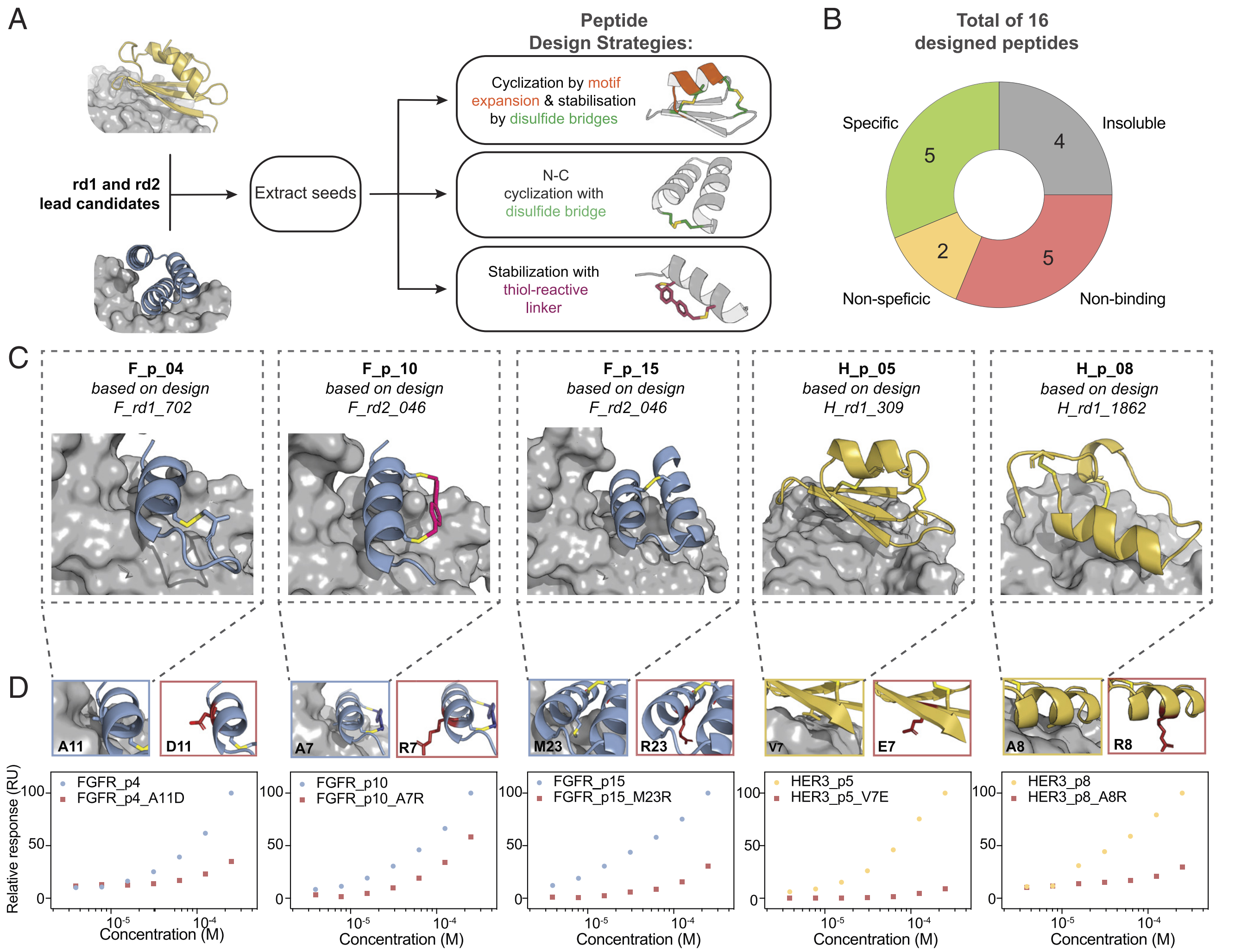

Fig. 5. Experimental characterization of stabilized peptides derived from de novo designed mini binders. (*A*) Overview of the employed peptide design strategies. The interface binding motifs from the experimentally validated rd1 and rd2 binders were extracted and used to design cyclized peptides. Three types of designs were generated, cyclic peptides with disulfides, peptides with a single disulfide, and peptides stabilized with thiol-reactive linkers. (*B*) Overview of properties of experimentally tested peptides. From a total of 16 synthesized peptides, four were insoluble, five did not show any binding to their target protein by SPR, four showed non target-specific binding, and five were target- and mode-specific binders. (*C*) Computational models of the five experimentally validated peptides targeting FGFR2 (blue) HER3 (yellow). (*D*) Affinity and specificity measurements of the five lead peptide candidates were performed using SPR. Maximum response at given concentrations is shown, highlighting specific peptide-target binding (FGFR2:blue, HER3:yellow) and their corresponding interface KO mutant (red), which exhibit reduced binding signal, thus demonstrating the peptides' target specificity.

targets, this represents, a large-scale structural survey of binder-accessible sites across the surfaceome, providing a framework for prioritizing tractable regions for protein engineering and therapeutic discovery.

To follow up on the prediction of binding sites across the surfaceome, we utilized a large library of 640,000 protein seeds to provide bona fide motifs to engage the targets of interest. Our seeds aimed to serve as anchoring motifs to guide design efforts on plausible binding geometries rather than final high-confidence solutions. This effort, equivalent to conducting 3 billion protein–protein docking and scoring simulations, provided detailed insights into the targetability of each predicted binding interface and identified a set of potential binding seeds to support future target-specific protein design initiatives. This depth of structural exploration is out of reach to other design methods relying on diffusion (9) or structure prediction networks (25). To experimentally validate the practical utility and applicability of the predicted target sites and selected seeds, we designed binders to target three proteins using two modalities, minibinders and stabilized peptides, leveraging the initial seed positioning while optimizing the binding interface with either Rosetta or ProteinMPNN. Minibinders demonstrated superior experimental successes compared to stabilized peptides yielding nanomolar to low micromolar binders against all selected targets.

Several recent efforts have focused on de novo protein binder design (9, 10, 25, 26). Having a defined target site on a protein of interest, both physics-based (5) and deep learning methods such as Rfdiffusion (9) or BindCraft (25) can provide unconditional designs with various success rates. Despite this progress, challenges remain in searching the vast space of possible binding modes and accurately identifying binding sites with high potential for targetability. Many current approaches lack for instance prior information about binding site properties or structural templates, leading to suboptimal starting points and potentially low experimental success rates. One way to address this issue is by leveraging high-quality preliminary data, including detailed structural information about binding sites and curated initial binder seeds which MaSIF can leverage at a proteome-wide scale. As demonstrated in the current study, such data can serve as a foundation to reduce the design space, improve prediction accuracy, and increase the likelihood of successful experimental outcomes. In this work, we also explored how to best use machine learning tools to enhance binding activity of designed binders, contributing to the general advancement of the protein design field for the engineering of molecular recognition. The identified binding sites, and corresponding seeds can also serve as starting points for diffusion-based methods such as RFdiffusion.

While our framework enables large-scale exploration of targetable sites, the binders generated in this study generally exhibit low affinities requiring further optimization to achieve higher binding affinities. In particular, the peptide designs have yet to reach optimal affinities, but they serve as an important proof of concept, demonstrating that our seeds can serve as starting points for the design of molecular formats beyond protein binders.

Furthermore, we introduced SURFACE-Bind (11), a first-of-its-kind large-scale interactive webserver, which allows the user to explore the analyzed target proteins and their top-scoring seeds for each predicted binding site, making it a valuable resource for biologically driven questions as well as for drug discovery efforts. This platform facilitates effortless browsing and data retrieval for specific target proteins. While this study focused on the human surfaceome, the database can be expanded to include surface proteins from other organisms, fostering advancements in protein binder design across diverse research efforts, including those with therapeutic applications.

Methods

Computational Workflow. The surfaceome database was collected from SURFY (2), where 2,886 proteins were predicted in silico to be on the surface of human cells. Corresponding AF2 models for each entry were obtained and filtered with the criteria of pLDDT > 0.6. MaSIF-site (13) was then used to predict the binding sites for all proteins and the results were clustered using DBSCAN, employing 0.7 as threshold for a point to be considered as a binding site. As each atom was sampled by 10 points, we discarded binding sites with a number of points <200 (1 to 3 residues coverage) and those located in TM domains, resulting in a total number of 4,500 binding sites. The following geometric and chemical features of each predicted binding site were computed: shape index, curvature (concave, convex, flat), area in A^2 (geometrical), sequence/chemical composition, SASA, and the hydrophobicity scale of Kyte and Doolittle (chemical). The bound state scores were calculated per seed through the "binding seed identification" workflow. All scripts and notebooks to run the workflow are available at: https://github.com/hamedkhakzad/SURFACE-Bind.

Binding Seed Identification. The fingerprint of the 12 Å (geodesic radius) patches found at the predicted interface of each target site was used to find complementary fingerprints in the MaSIF-seed database. The database consists of ~640,000 continuous structural fragments (seeds), amounting to 402 million surface patches/fingerprints (10). The MaSIF-search algorithm was trained to make patch fingerprints similar for interacting patches and dissimilar for noninteracting patches. As described previously (10), seeds are selected based on a high interface propensity score and a low fingerprint distance (Euclidean distance between target and seed fingerprint). In a second-stage alignment and scoring using the RANSAC algorithm, seeds were selected based on the interface postalignment (IPA) score. The binding seed identification was applied to all 4,500 sites, performing nearly 3 billion pairwise docking runs to obtain the final list of seeds per target site.

Seed and Interface Refinement. As done previously (10) to optimize the binding energy of seeds in silico, a FastDesign protocol in Rosetta (27) was employed, incorporating a penalty for buried unsatisfied polar atoms in the scoring function (28). Optimized seeds were selected based on several criteria: computed binding energy (ddG), shape complementarity, the number of interface hydrogen bonds, the number of buried unsatisfied polar atoms, and the number of atoms in contact with the small molecule. Beta sheet-based motifs making more than 33% contact with the target complex using loop regions were discarded. Additionally, the uniqueness of each seed was assessed through pairwise alignment of hotspot residues. For seeds showing more than 70% hotspot identity with another seed, only the one with the best surface-normalized ddG was retained.

Seed Grafting. Selected seeds were then grafted using the Rosetta MotifGraft protocol (29) to stabilize the binding motif and enhance contact with the target complex as done in (10). Each seed was matched with a database of approximately 6,500 small protein scaffolds (under 90 amino acids) derived from small globular monomeric proteins in the Protein Data Bank (PDB) (30) and four miniprotein databases that had been experimentally validated (31–34). Seeds were trimmed to the minimum number of residues making contact with the target, and loop motifs were removed from beta sheet-based seeds to optimize grafting success. After grafting, scaffolds underwent sequence optimization using a similar FastDesign protocol in Rosetta as for the seed refinement. Final designs were selected based on computed binding energy (ddG), shape complementarity, the number of interface hydrogen bonds, and the count of buried unsatisfied polar atoms.

Structural and Sequence Similarity Calculations. To calculate the structural similarity between the AF2 prediction and the initial designs, the C_α rmsd was calculated as the mean Euclidean distance between the aligned structures via Biopython package (35). The sequence similarity between two given sequences was calculated using the Biotite Python package (v. 0.41) (35, 36).

Sequence Optimization Using ProteinMPNN. We used a ProteinMPNN model with soluble MPNN ($MPNN_{sol}$) weights, trained on a dataset comprising only soluble proteins (37). The designs generated in round 1 were utilized as inputs to ProteinMPNN. Sequence optimization was performed in two settings: i) Fixed interface, where interacting residues of the designs within 3.5 Å from the target interface were kept fixed while ProteinMPNN redesigns the scaffold residues, and ii) Full redesign, where all residues were optimized. For each design, 50 sequences were generated, 25 sequences with each setting using the soluble model weights and sampling temperature of 0.1. From each setting, the top five unique sequences per design, based on ProteinMPNN global score, were forwarded to AF2 for subsequent filtering steps.

Design Filtering Using AF2. The in silico foldability of the ProteinMPNN-generated sequences was assessed using AF2 via ColabFold (v1.5.2) (38). Predictions were performed with three recycles and in single sequence mode (no multiple sequence alignments were provided). The AF2 scores from the five prediction models were averaged and used to filter the optimized sequences based on the following criteria: 1) pLDDT $\geqq$ 90, 2) pTM $\geqq$ 0.7, and 3) C_α rmsd to design model $\leqq$ 1.5 Å. For HER3, to increase the number of experimentally tested designs, the pLDDT threshold was lowered ($\geqq$80) and the pTM filter was not used.

The binder–target complex interface of the passing designs was further evaluated using complex prediction performed with the AF2 monomer model via ColabDesign (v1.1.1) (39). Custom templates of the target–binder complex were provided for these predictions. The designs were subsequently filtered based on their ipTM $\geqq$ 0.7. Finally, a selection of successful designs was chosen for experimental testing.

Computational Performance Benchmark. The computational success rate of our seed-centered design strategy in generating potential binders satisfying the AF2 filtering criteria was compared to the state-of-the-art generative diffusion-based method RFdiffusion (v1.1.0). We generated 2,570 binder backbones per target against the same sites identified by MaSIF-site, by providing the target interface residues within 3.5 Å of our successful round two binders as hotspots. The generated binder lengths ranged from 30 to 88 residues to match our scaffold lengths distribution. A backbone radius of gyration (monomer_ROG) potential was introduced to favor the generation of compact folds. Additionally, a length-dependent radius of gyration filter was applied to exclude trivial helical structures and elongated backbone ($R_g < 4N3/5 + 9$) with (N = residue-number) based on ref. 40. For approximately 2,000 filtered backbones per target, five sequences were generated using ProteinMPNN in Full redesign setting as described in (*Methods: Sequence Optimization Using ProteinMPNN*). The generated sequences were evaluated for in silico foldability and interface quality using AF2 as described in (*Methods: Design Filtering Using AF2*). The computational success rate is defined as the number of designs passing the thresholds (Monomer: AF2 pLDDT $\geqq$ 90, AF2 pTM $\geqq$ 0.7, and C_α rmsd to design model $\leqq$ 1.5 Å; Complex: AF2 ipTM $\geqq$ 0.7).

Library Preparation. For each target complex—namely FGFR2, IFNAR2, and HER3—around 2,000 designs were reverse-translated into DNA and purchased from Twist Bioscience as oligo pools with 18 base pairs homology overhangs with the final vector. These oligo pools were subjected to two rounds of PCR: first, to amplify the library using the 18 base pairs overhangs, and second, to add 45 base pairs homology with the yeast display vector (with an annealing temperature of 57.5 °C for 30 s, and an extension at 72 °C for 1 min, over 15 cycles). The amplified inserts and linearized pCTcon2 vector (with C-terminal HA-tagged) were used to transform EBY-100 yeast by electroporation, following previously established protocols (41). The transformed yeast cells were cultured in minimal glucose (SDCAA) medium at 30 °C and then induced overnight in minimal galactose (SGCAA) medium before sorting.

Flow Cytometry Analysis and Sorting. Induced yeast cells were rinsed with PBS containing 0.1% BSA and then labeled with the target binding molecule (with Fc-fusion) for 2 h at 4 °C. The cells were subsequently pelleted and labeled with a FITC-conjugated goat anti-HA tag antibody (Bethyl; ref: A190-138F; display tag; 1:100 dilution) and a PE-conjugated goat anti-human Fc antibody (Invitrogen; ref: 12-4317-87; binding tag; 1:100 dilution) for 30 min at 4 °C. Finally, pelleted cells were resuspended in an appropriate PBS/BSA buffer volume and analyzed using a Gallios flow cytometer (Beckman Coulter) or sorted with a Sony SH800 cell sorter. Data acquisition was performed using the Kaluza software (Beckman Coulter, v.1.1.20388.18228) for flow cytometry and the LE-SH800SZFCPL Cell Sorter software (Sony, v.2.1.5) for fluorescence-activating cell sorting. Each designed library was sorted into a binding and a nonbinding population. The flow cytometry data were then analyzed with the FlowJo software (BD Biosciences, v.10.8.1) and steady state responses and dissociation constants were plotted using a nonlinear four-parameter curve fitting analysis. For the competition assays, the same procedure was performed but before addition to yeast the target protein was first preincubated with an 1:10 molar excess of the competitor.

Library Sequencing. Sorted yeast were cultured, and plasmids encoding protein designs were extracted using the Zymoprep Yeast Plasmid Miniprep II kit (Zymo Research) according to the manufacturer's instructions. The gene of interest was then amplified by PCR using vector-specific primers flanking the protein design gene. A second PCR was conducted to add Illumina adapters and Nextera barcodes. The PCR product was then desalted and purified using the Qiaquick PCR purification kit (Qiagen). Next-generation sequencing was performed using the Illumina MiSeq system with 500 cycles. Approximately 0.8 to 1.2 million reads per sample were obtained, translated into the appropriate reading frame, and matched with the expected input sequences from the libraries. The enrichment of each design was calculated by dividing the counts in the binding population with those in the nonbinding populations. Finally, hits were identified if the enrichment in the binding population was more than 10-fold greater than in the nonbinding population, and the number of counts in the binding population exceeded 10,000.

Yeast Surface Display of Single Designs. Genes encoding single designs were obtained from Twist Bioscience, featuring ~25 base pairs homology overhangs for cloning. Each design was inserted into pCTcon2 plasmids (with C-terminal HA-tagged) via Gibson assembly and transformed into HB101 competent bacteria for DNA production. After purification and sequence verification by Sanger sequencing, the DNA was used to transform competent EBY-100 yeast with the Frozen-EZ Yeast Transformation II Kit (Zymo Research). Transformed yeast cells were cultured in minimal glucose (SDCAA) medium at 30 °C and then induced overnight in minimal galactose (SGCAA) medium before undergoing flow cytometry analysis.

Protein Expression and Purification. Protein sequences used in this work are listed in *SI Appendix*, Table S7. Genes encoding the 6xHis-tagged and/or human Fc-tagged proteins were obtained from Twist Bioscience, cloned into pET-11 (bacterial vector) or pHLSec (mammalian vector) using Gibson assembly, and transformed into HB101 bacteria. Plasmids were extracted using a GeneJET plasmid Miniprep kit (ThermoFisher) for bacterial vectors or a Plasmid Plus Midi Kit (Qiagen) for mammalian vectors and verified by Sanger sequencing. Proteins were then purified using either bacterial or mammalian expression systems.

For mammalian expression, the Expi293 system (ThermoFisher; ref: A14635) was used. Supernatants were collected after 6 d, filtered, and purified as described below. For bacterial expression, BL21(DE3) or T7 Express Competent *E. coli* were transformed with the plasmid and grown overnight in a preculture. Precultures were then inoculated 1:50 in 500 mL medium and incubated at 37 °C until reaching an OD_{600} of ~0.7. The cultures were induced with 1 mM IPTG and incubated overnight at 18 to 20 °C. Cells were harvested by centrifugation at 4,000 g for 10 min, resuspended in lysis buffer (50 mM Tris, pH 7.5, 500 mM NaCl, 5% glycerol, 1 mg/mL lysozyme, 1 mM PMSF, and 1 μg/mL DNase), and lysed by sonication. Lysates were clarified by centrifugation at 30,000 g for 30 min and filtered. All 6xHis-tagged proteins were purified using the ÄKTA pure system (GE Healthcare) with a Ni-NTA HisTrap affinity column, followed by size exclusion chromatography on a Superdex HiLoad 16/600 75 pg or 200 pg column, depending on the protein size. Proteins were concentrated in PBS as the final buffer.

Surface Plasmon Resonance Measurement (SPR). Affinity measurements were conducted using a Biacore 8K system (GE Healthcare, software v4.0.8.19879) with HBS-EP+ as the running buffer (10 mM HEPES at pH 7.4, 150 mM NaCl, 3 mM EDTA, 0.005% v/v Surfactant P20, GE Healthcare). Proteins were immobilized on a CM5 chip (GE Healthcare #29104988) via amine coupling to achieve 500 to 1,500 response units (RU). Analytes were injected in serial dilutions using the running buffer, with a flow rate of 30 μL/min for a contact time of 120 s, followed by a dissociation time of 400 s. The SPR data were analyzed by steady-state affinity mode, reporting the RU normalized to the highest RU reached for each concentration. Data were fit with a 1:1 Langmuir binding model within the Biacore 8K analysis software (Cytiva, v.4.0.8.19879).

Size-Exclusion Chromatography–Multi-Angle Light Scattering (SEC-MALS). To determine the molecular weight of the purified designs, SEC-MALS on a miniDAWN TREOS instrument (Wyatt) was utilized. The samples, at a concentration of approximately 1 mg/mL in PBS (pH 7.4), were injected into a Superdex 75 10/300 GL column (GE Healthcare) with a flow rate of 0.5 mL/min. Signals for UV_{280}, refractive index (dRI), and light scattering were recorded. The molecular weight was calculated using the ASTRA software (version 6.1, Wyatt).

Circular Dichroism. Far-ultraviolet circular dichroism spectra were recorded using a Chirascan spectrometer (Applied Photophysics). Protein samples were diluted in PBS to a concentration of 300 μg/mL and placed in a 1 mm path-length cuvette. Wavelengths between 200 and 250 nm were scanned at a speed of 20 nm/min with a response time of 0.125 s, with all spectra corrected for buffer absorption. Temperature ramping melts were conducted from 20 to 90 °C at a rate of 2 °C/min. Thermal denaturation curves were plotted by monitoring changes in ellipticity at the global minimum, and the melting temperature (T_m) was determined by fitting the data to a sigmoid curve equation using GraphPad Prism.

Stapling of Peptides with a Chemical Linker. The peptides were chemically synthesized using solid-phase peptide synthesis and the chemical linker was introduced following the procedure described by Diderich et. al. (42) with some modifications. The thiol-reactive linker employed in this study is 4,4′-Bis(bromomethyl)biphenyl. The synthesized peptide was incubated with the chemical linker in the presence of a reducing agent (TCEP) with 1:1.5:3 molar ratios, respectively. The reaction was incubated for 120 min at 25 °C and 1,000 rpm in a thermomixer. Finally, the stapled peptide was purified using reversed-phase high-performance liquid chromatography (RP-HPLC) and lyophilized until further use. This procedure was performed by GL biochem (Shanghai), Ltd., where the peptides were purchased from.

Generation of the Disulfide Fragment Hashtable. N-to-C terminus disulfide monocycles ranging from 6 to 15 mers were generated using the kinematic closures algorithm in Rosetta (32). Briefly, starting with polyglycine with N and C terminal cysteines, backbone torsions were biasedly sampled based on rama probability. Solutions with ideal disulfide chi angles are clustered based on backbone torsion bin strings (43). To generate the hashtable, sliding windows containing at least three residues containing both cysteines were taken. The hash key is the affine transform between N-C_α-C local frames at perspective ends and the value corresponds to the internal degree of freedom of the peptide fragments (44).

Cyclotide Design of Disulfide-Dense Cyclized Peptides. Binding motifs were extracted from experimentally validated round 1 (rd1) designs as. A disulfide-anchored loop was added to the selected motifs through disulfide fragment lookup using the hashtable generated from disulfide monocycles. During the lookup, one external N-C_α-C and one internal N-C_α-C frame were sampled to identify potential disulfide placements while extending the motif. Next, the motifs were stabilized through incorporation into larger—up to 38 residues—cyclic peptides employing RFDiffusion (9). Briefly, an artificial chain break within the linear peptide was defined, and standard RFDiffusion structure generation was used to cyclize the peptide termini. To further enhance the peptide's stability, a second disulfide bond was introduced via Rosetta (32) when feasible. The cyclized peptides were evaluated using an AF2-based cyclic peptide and protein structural prediction network (45). Predictions were filtered based on the following criteria: 1) pLDDT > 80, 2) ipAE (interaction predicted alignment error) < 5, and 3) C_α rmsd to design model <2.5 Å. Filtered designs were then assessed using MD simulations for their fold stability.

MD Simulation Protocol. MD simulations were performed in explicit TIP3P solvent using the GROMACS 2023 package (46) and the RSFF2 force field (47). RSFF2 is an AMBER-derived force field which was parameterized using a coil library and shown to capture intrinsic conformational preferences of amino acids with high accuracy. Initial structure predictions for 29 representative designs were solvated in a cubic box of pre-equilibrated solvent, with a minimum distance between the peptide and the box boundaries set to 1.0 nm. The solvated structure was energy-minimized using the steepest descent algorithm, and then equilibrated in four stages: 50 ps in the canonical ensemble (NVT) at 300 K, 50 ps isothermal–isobaric ensemble (NPT) at 300 K and 1 bar, both with position-restrained heavy atoms; 100 ps NVT at 300 K, 100 ps NPT at 300 K and 1 bar, both unrestrained. An NPT production run was conducted for 250 ns, at 300 K and 1 bar, with a simulation time step of 2 fs and data recorded every 1 ps.

All simulations were performed using the standard leapfrog algorithm, with SETTLE used to maintain water geometry (48), and LINCS applied to constrain hydrogen-containing bonds to equilibrium lengths (49). Nonbonded interactions were treated explicitly up to 1.0 nm, with Coulombic interactions past this threshold computed via particle mesh Ewald summation, using a 0.12 nm Fourier spacing and cubic interpolation. Dispersion corrections for both energy and pressure were applied to the long-range van der Waals interactions. The V-rescale thermostat with a coupling time constant of 0.1 ps was used to control temperature, and the Parrinello–Rahman barostat with a coupling time constant of 2.0 ps and isothermal compressibility of 4.5×10^{-5} bar^{-1} was used to maintain pressure (50).

All systems were simulated with disulfide bonds replaced with free-sulfhydryl cysteines, and the pairwise distances for all possible pairs of cysteine sulfur atoms were recorded for each frame of the simulation. Pseudodisulfide frames were defined as all frames with any S–S distance lower than 4 Å. For each system, the overall population of such frames, as well as the relative population of all represented combinations of spatially proximate cysteine pairs, was recorded. Systems with a single dominant combination matching the original design were considered "more synthesizable" and proceeded to the next stage of testing, under the assertion that their conformational ensembles would minimize undesirable side products being formed during synthesis. In addition, to ensure that the stereochemistry of the disulfide bonds would match the original design, we also monitored the distribution of the torsion angles corresponding to the pseudodisulfides throughout the simulation trajectories. Any systems displaying multimodal distributions, or significant peaks not matching the originally predicted structure, were discarded at this stage, whereas the remaining systems proceeded to experimental evaluation.

Data, Materials, and Software Availability. All data, including comprehensive details of binding sites, unbound-state scores, predicted seeds, and bound-state scores, are freely accessible on Zenodo (10.5281/zenodo.15016859) (51). Additionally, the SURFACE-Bind web server (https://surface-bind.inria.fr/) (11) offers an interactive platform for exploring predicted binding sites and their corresponding seeds. The code and scripts for reproducing the results of this study are freely available online. MaSIF-site can be accessed at https://github.com/LPDI-EPFL/masif (52), while the seed search protocol, based on MaSIF-seed-search, is available at https://github.com/LPDI-EPFL/masif_seed (53). Additionally, all scripts for executing the workflow on a large scale, defining unbound-state scores for binding sites, analyzing the results, and optimizing binders resulting from MaSIF-seed-search can be found at https://github.com/hamedkhakzad/SURFACE-Bind (54). All other data are included in the manuscript and/or supporting information.

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Competing interest statement: T.-Y.Y., J.D., S.F., and C.Y. were previously or are currently employed by Novo Nordisk A/S. A.M. is now employed of bNovate Technologies SA. T.-Y.Y. is now employed by Amazon. S.F. is now employed by Novartis AG. All other authors declare no competing interests.

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